<u>Objective 1: Cost Optimization Research (Trip Cost Prediction/ Regression Model)</u>

<u>Introduction</u>

This research is to develop a predictive model for **Fishing Trip Operational Cost** with only considering the scenario that estimates total trip expenditure using only pre-departure variables such as forecasted weather, distance, and engine specifications to enable financial decision-making before a vessel leaves the harbor. This study hopes to provide ship owners with an accurate budgeting tool that does not rely on post-trip actuals. If the owner knows the tomorrow's weather and after filling out the pre-departure variables, the owner can accurately estimate the total operational cost of voyage before it begins.

<u>Preparing the data</u>

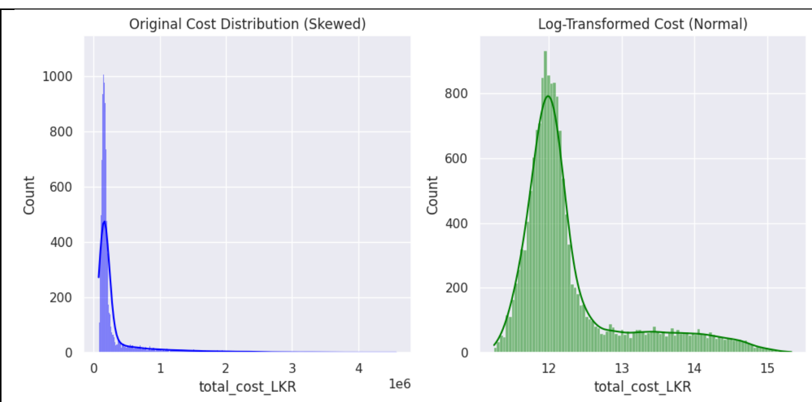
In the data preparation always, the model remained "blind" to any information gathered during the trip. This involve removing the cost components like fuel cost, food cost and catch weights which directly influence the target variable "Total Cost", forcing the model to relay on the weather, physical conditions of the vessel and the trip planning data such as the crew size, trip type and allowed off shore days. Categorical features were transformed using **One-Hot Encoding**, while numerical features were inspected for missing values. To handle the significant right-skewness of the cost data, a log-transformation was applied.

<u>Exploratory Data Analysis</u>

Exploratory Data Analysis was conducted with the normal target values and also the log transformed values. The clearer pictures were obtained for the log transformed plots because of the high skewness in the data. Therefore, the following plots are plotted with the log transformed target variable.

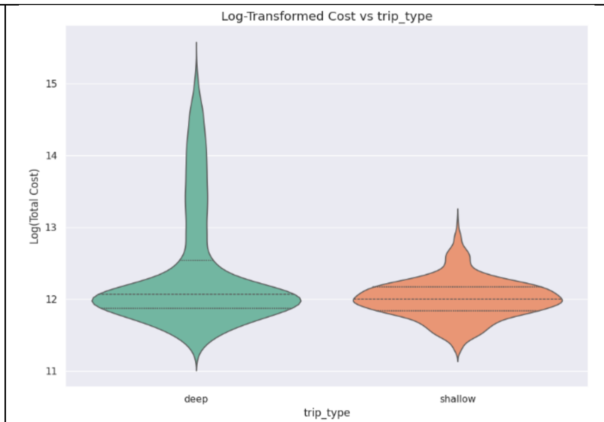
Log Transforming the price

The original cost data is extremely right-skewed, by shifting the distribution from right skewed tail towards a normal curve the transform says that the model errors are consistent across both short distance and long-distance voyages.



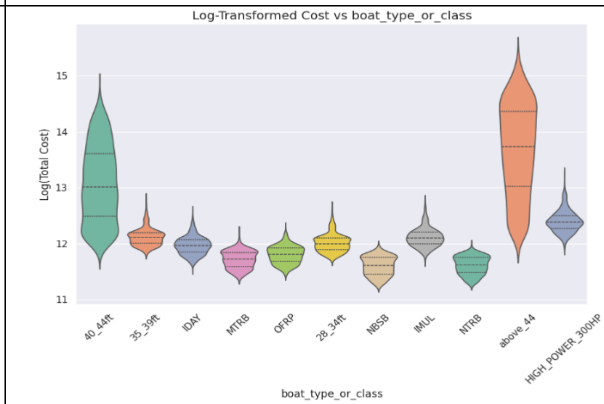
Log transformed cost vs trip type

After log transforming the distribution become clearer, reducing skewness and making central tendencies clear for the both trip types. Overall, the plot suggest that the deeper trips are generally more expensive.



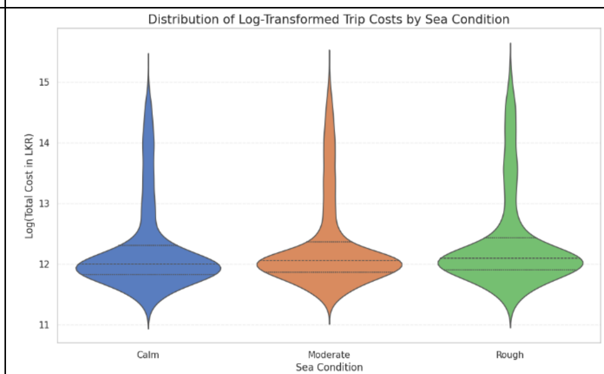
Log transformed cost vs boat type or class

The violin plot shows that the mid-sized categories such as 40–44 and HIGH POWER >300HP also have relatively higher costs compared to smaller classes like NTRB, NBSB, and MTBB, which cluster at lower log-cost levels. Overall, cost tends to increase with boat size/class, and variability becomes larger for higher-end boat categories.



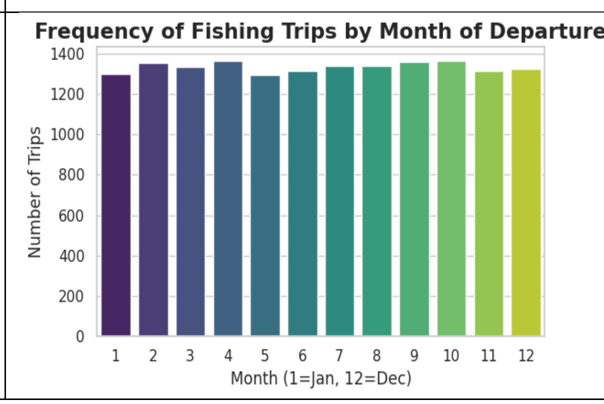
Log transformed TRIP Costs by Sea Condition

As the results in the plot which has plotted with the price itself didn't give a clear picture, here the log transformed price was used. The median costs slightly increase from calm to rough conditions indicating higher typical expenses in harsher seas. Rough conditions also show wider spread, suggesting greater variability and risk in trip costs under rough seas.



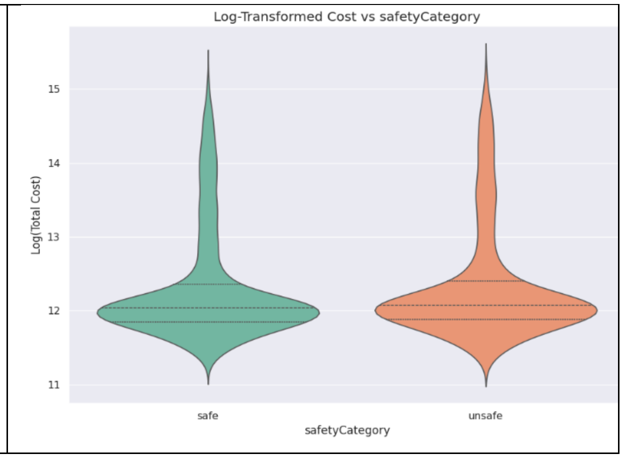
Frequency of Fishing Trips by Month of Departure

The bar chart shows that fishing trips are evenly distributed across all months, with only small fluctuations throughout the year. There are slight increases in trips during mid to late months, suggesting a seasonal peak. Overall, no none of the months dominate indicating consistent fishing activity year-round.



Log Transformed Cost vs Safety Category

The violin plot shows that both safe and unsafe categories have similar central log-cost values, clustered around 12. The unsafe category has a wider spread and higher upper tail indicating that greater variability and more high-cost trips. This suggests that while average costs are comparable, unsafe trips are more likely to incur extreme expenses.



Multicollinearity Assessment and the statistical tests

Before modeling VIF values were checked to clarify whether the multicollinearity exist. Furthermore, non-parametric statistical tests were used to validate the importance of specific categories. **Mann-Whitney U Test** confirmed that there is a statistically significant difference in costs between the trips labeled as “Safe” and “Unsafe”. Additionally, a **Chi-square test of Independence** performed between weather conditions and safety categories. The resulting p-value proved that weather and safety are dependent which means the bad weather explain that it’s not safe. Following are the tables values.

- Variance Inflation Factor (VIF)

Feature	VIF
boat_length_ft	24.217020
engine_hp	20.836657
crew_size	3.633029
allowedOffshoreDays	1.763567
distance_km	1.225762
wave_m	1.002121
wind_kph	1.002063

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Mann-Whitney U statistic: 27394993.5000
P-value: 6.862e-08
Conclusion: Reject the Null Hypothesis (p < 0.05).
There IS a statistically significant difference in the distribution of costs between Safe and Unsafe trips.
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--- Chi-Square: Link between Weather and Safety ---
Chi-Square Statistic: 9365.6536
P-value: 0.0000

Conclusion: Weather and Safety are DEPENDENT. Rough weather is statistically linked to safety risks.
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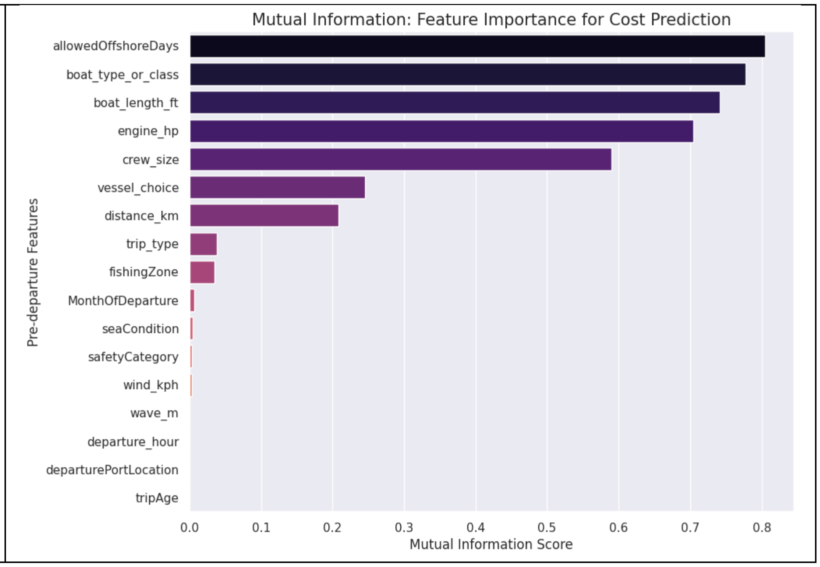
Mutual Information

The plot shows that allowed Off shore Days, boat type or class, and boat length are the most informative features for predicting cost, followed by engine hp and crew size. Mid-level contributors include vessel choice and distance(km), while variables like trip type and fishing Zone add only minor information.



Data Preprocessing

All the numerical variables were scaled during the modeling process and all the categorical variables were encoded using one-hot encoding.

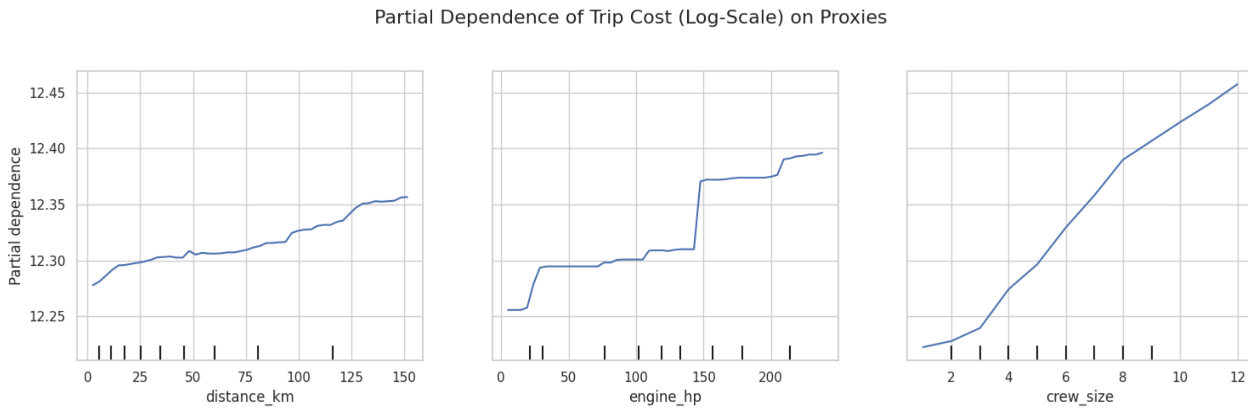
Model Development and evaluation

Five distinct models were developed and checked for the R squared values. At first models were fitted using the original Total Cost variable. The RMSE scores were very high due to the large scale of data. Therefore, models were again fitted using the log-transformed Total Cost variable. Then the loss functions were quit low and the R squared values were quite high. Following are the results after training the model using the log transformed target variable.

Model	Train R2	Test R2	Train MAE (LKR)	Test MAE (LKR)	Test MSE (LKR)	Test RMSE (LKR)
XGBoost	0.984247	0.979215	20223.097091	27121.307335	4.238163e+09	65101.177062
Random Forest	0.996843	0.978300	10745.076769	29132.641639	4.951544e+09	70367.209700
Ridge	0.963531	0.964419	48257.039038	49807.641487	1.446451e+10	120268.482688
Elastic Net	0.958727	0.959585	54834.954527	56399.431524	1.968309e+10	140296.441065
Lasso	0.955897	0.956720	57086.023712	58350.201481	2.067825e+10	143799.351322

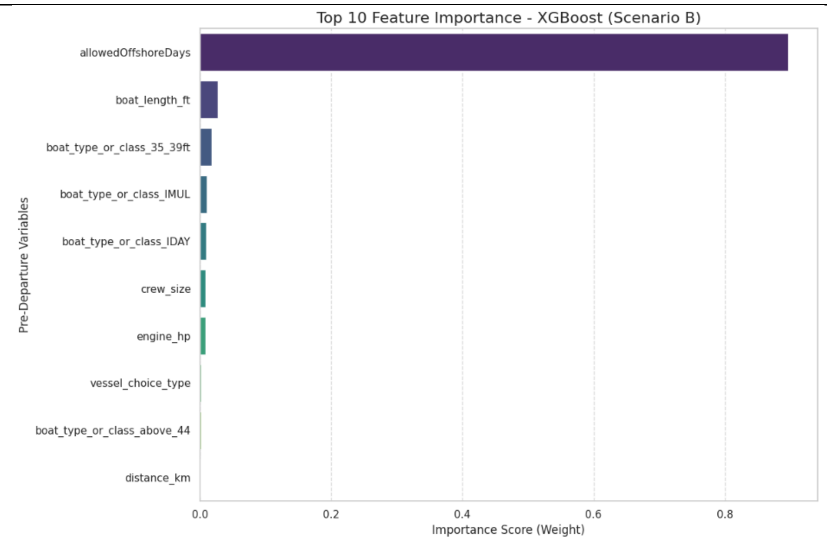
It has one of the highest tests R^2 value (0.9792) and the lowest test MAE (27,121 LKR) and the lowest RMSE (65,101 LKR), meaning it explains the most variance and make the smallest prediction errors on unseen data. Random Forest also close, but overall XGBoost still performs slightly better overall.

Following are the partial dependencies of the mostly influenced variables. It says when increasing the distance after keeping every variable constant, the total price is getting increased. Same scenario happens with engine_hp and crew size.



Feature Importance

Cost is being driven mainly by how many offshore days are allowed, with vessel characteristics and the crew size. The similar results were gained from the mutual information plot too. Therefore, the model is relying primarily on offshore permission days to predict cost, and the rest of the features play only a minor supporting role.



Conclusion

This research shows that it is possible to estimate fishing trip costs before departure with high accuracy. Among all the model tested, XGBoost performed best. The study also identified “Weather”, “Labor” and “Zone” have a major impact on costs. The results will help boat owners to decide about the crew size and the departure timing, to lower the cost which leads for mor sustainable fishing operations.